

District 7 Model Comparison - Geospatial Data: 2023 - 2025 Sales

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

15 July 2026

Contents

Model I	1
Model II	1
Model III	2
Model IV	2
Model V	3
Model VI	3
Model Comparison	4
Prediction Testing	5

Model I

Table 1: Model Metrics: LN-SPPSF Model Comparison

Metric Source	n	df_resid	df_model	r	R ²	Adj. R ²	SEE	f_stat	AIC	BIC
Control	1386	1370	15	0.3663	0.1341	0.1247	0.4521	14.1499	1750	1839
+ Geospatial	1385	1368	16	0.4329	0.1874	0.1779	0.4383	19.7205	1664	1758
% Difference	-0.07%	-0.15%	6.67%	18.20%	39.72%	42.72%	-3.06%	39.37%	-4.91%	-4.40%

Table 2: Ratio Metrics: LN-SPPSF Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9865	1.1083	0.9089	1.2193	40.0356	-0.9777	17.60%	32.61%	73.45%
+ Geospatial	0.9747	1.1022	0.9145	1.2053	39.4801	-0.8282	17.98%	33.94%	75.23%
% Difference	-1.20%	-0.55%	0.61%	-1.15%	-1.39%	15.30%	2.12%	4.06%	2.43%

Model II

Table 3: Model Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	n	df_resid	df_model	r	R ²	Adj. R ²	SEE	f_stat	AIC	BIC
Control	1806	1790	15	0.4647	0.2159	0.2094	0.4680	32.8677	2400	2493
+ Geospatial	1805	1787	17	0.4913	0.2414	0.2342	0.4606	33.4446	2343	2448
% Difference	-0.06%	-0.17%	13.33%	5.72%	11.77%	11.83%	-1.57%	1.76%	-2.38%	-1.81%

Table 4: Ratio Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9817	1.1169	0.8979	1.2439	41.4895	-0.7183	16.00%	33.17%	73.37%
+ Geospatial	0.9756	1.1135	0.9010	1.2358	41.1555	-0.6681	16.01%	33.07%	75.12%
% Difference	-0.62%	-0.31%	0.34%	-0.65%	-0.81%	7.00%	0.06%	-0.28%	2.40%

Model III

Table 5: Model Metrics: LN-PRICE Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1806	1779	26	0.5804	0.3368	0.3272	0.4317	34.7546	2119	2273
+ Geospatial	1805	1774	30	0.5850	0.3422	0.3311	0.4305	30.7664	2112	2288
% Difference	-0.06%	-0.28%	15.38%	0.80%	1.60%	1.21%	-0.29%	-11.48%	-0.33%	0.66%

Table 6: Ratio Metrics: LN-PRICE Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9855	1.0974	0.9137	1.2011	37.7055	-0.4988	18.27%	34.83%	77.13%
+ Geospatial	0.9860	1.0962	0.9142	1.1991	37.5873	-0.4886	17.12%	35.29%	77.34%
% Difference	0.05%	-0.11%	0.06%	-0.17%	-0.31%	2.04%	-6.31%	1.33%	0.27%

Model IV

Table 7: Model Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1648	1621	26	0.6274	0.3936	0.3839	0.3748	40.4716	1470	1622
+ Geospatial	1629	1600	28	0.6434	0.4139	0.4037	0.3678	40.3584	1395	1557
% Difference	-1.15%	-1.30%	7.69%	2.55%	5.16%	5.15%	-1.85%	-0.28%	-5.10%	-4.01%

Table 8: Ratio Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9961	1.0717	0.9358	1.1452	32.2970	-0.4079	20.08%	38.90%	81.37%
+ Geospatial	0.9912	1.0690	0.9374	1.1404	31.8994	-0.3789	19.03%	38.98%	82.44%
% Difference	-0.49%	-0.26%	0.17%	-0.43%	-1.23%	7.11%	-5.25%	0.22%	1.32%

Model V

Table 9: Model Metrics: LN-TASP Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	1648	1622	25	0.6063	0.3676	0.3578	0.3751	37.7124	1472	1618
+ Geospatial	1629	1599	29	0.6245	0.3899	0.3789	0.3680	35.2431	1398	1565
% Difference	-1.15%	-1.42%	16.00%	2.99%	6.08%	5.88%	-1.87%	-6.55%	-5.03%	-3.28%

Table 10: Ratio Metrics: LN-TASP Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9995	1.0719	0.9353	1.1460	32.2287	-0.4395	20.21%	38.53%	81.01%
+ Geospatial	0.9933	1.0691	0.9370	1.1410	31.8140	-0.4058	18.91%	39.53%	82.50%
% Difference	-0.63%	-0.26%	0.18%	-0.44%	-1.29%	7.68%	-6.43%	2.60%	1.85%

Model VI

Table 11: Model Metrics: LN-TASP Model II Comparison

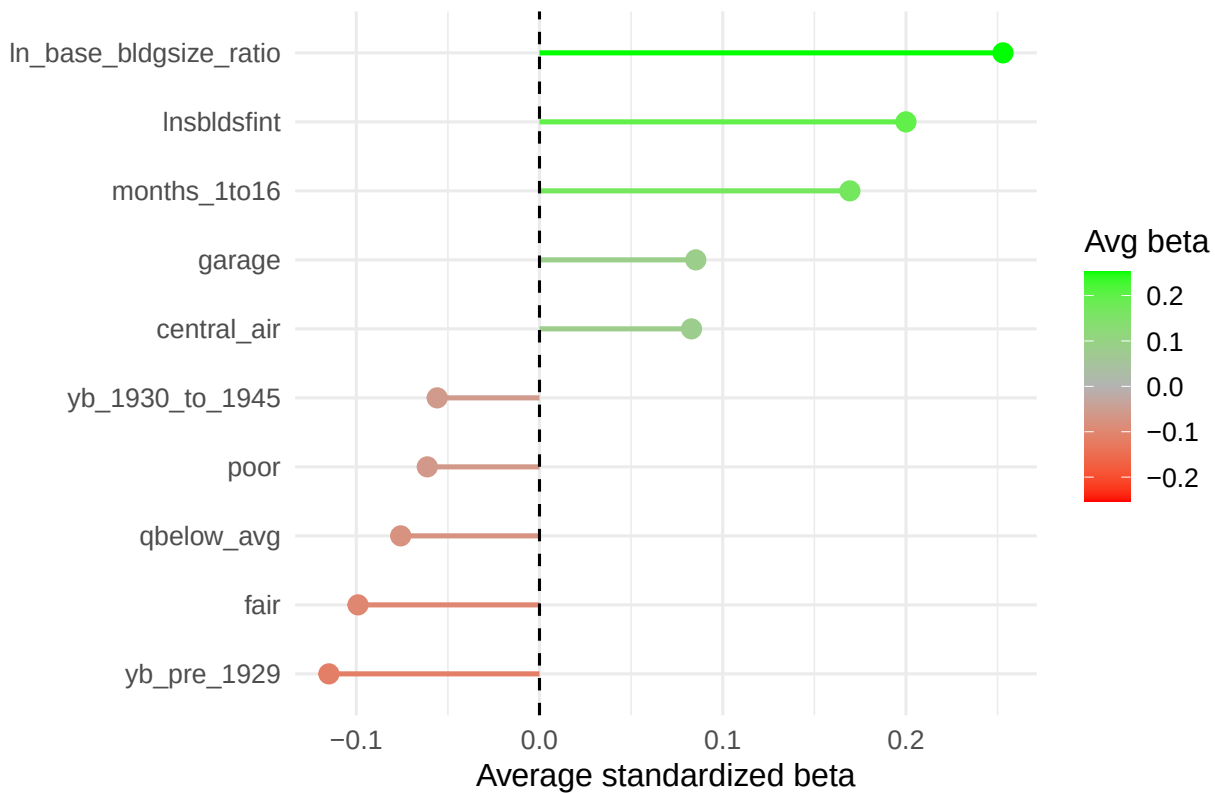
Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	2666	2634	31	0.5784	0.3346	0.3268	0.4434	42.7269	3262	3457
+ Geospatial	2665	2631	33	0.5825	0.3394	0.3311	0.4420	40.9547	3247	3453
% Difference	-0.04%	-0.11%	6.45%	0.71%	1.42%	1.32%	-0.31%	-4.15%	-0.46%	-0.12%

Table 12: Ratio Metrics: LN-TASP Model II Comparison

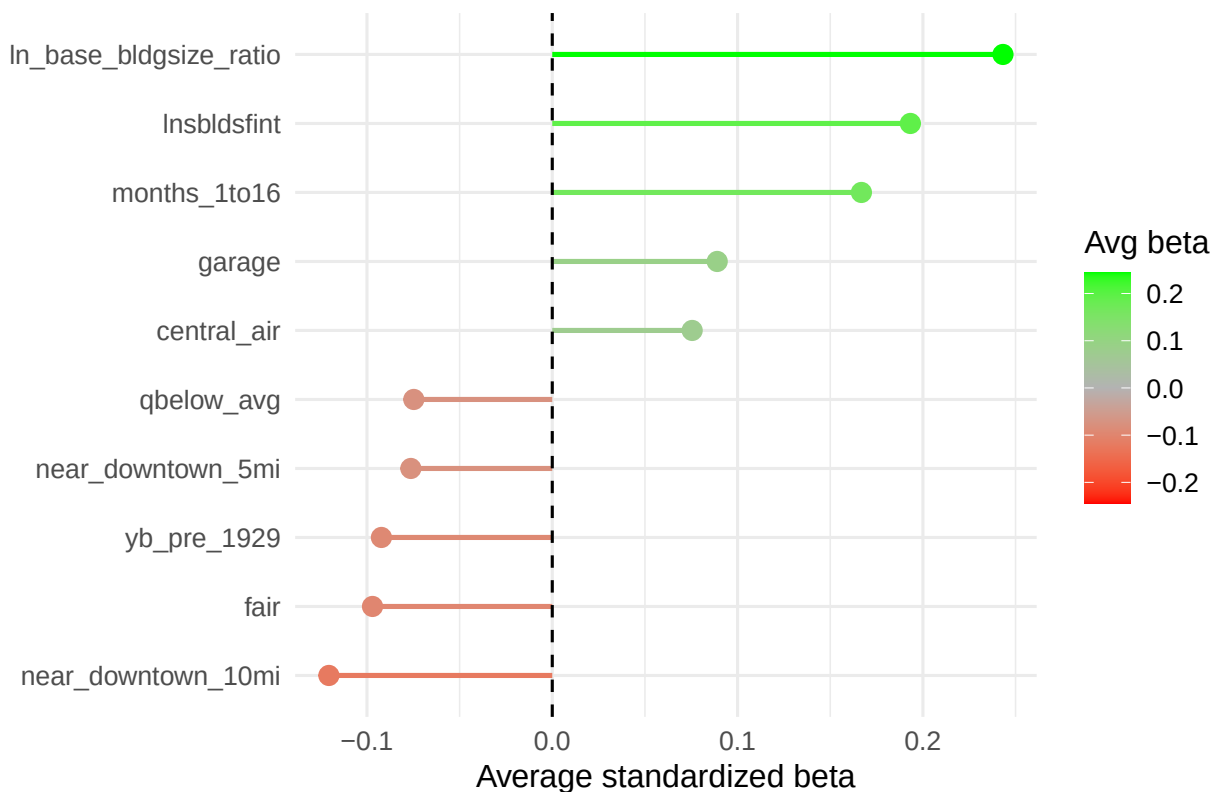
Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9872	1.1026	0.9118	1.2092	38.7457	-0.5248	17.10%	34.02%	75.62%
+ Geospatial	0.9805	1.1022	0.9122	1.2083	38.9168	-0.5183	17.19%	34.56%	75.76%
% Difference	-0.68%	-0.03%	0.04%	-0.08%	0.44%	1.23%	0.48%	1.58%	0.19%

Model Comparison

District 7 | Avg Std Beta across Initial Models



District 7 | Avg Std Beta across Geospatial Models



Prediction Testing